Food Delivery System

EFFICIENTLY MANAGING FOOD DELIVERY OPERATIONS THROUGH A STRUCTURED DATABASE SYSTEM.

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Problem Statement:

In food delivery systems, managing data like customers, restaurants, orders and payments can become complex. Without a proper database, it leads to errors and duplication. So, a structured database system is needed to manage this data efficiently.

PROJECT OVERVIEW:

Key Features:

- Customer registration and data management
- Restaurant and menu management
- Order placement
- Payment handling
- Delivery partner management
- Customer reviews and ratings

Purpose:

- Organize data in a structured format
- Reduce data redundancy
- Ensure data consistency
- Enable efficient data retrieval

SYSTEM ARCHITECTURE:

Presentation Layer(Frontend)

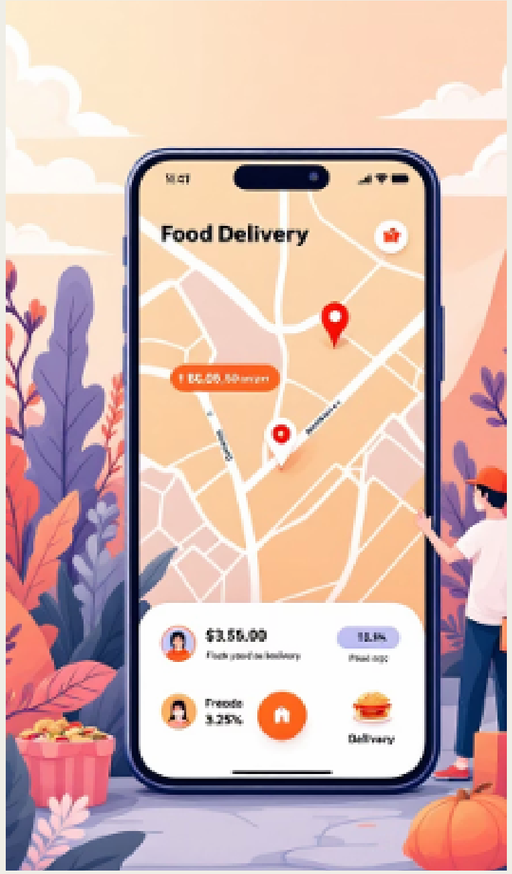
- User interface (website)
- Allows users to:
  - -View restaurants
  - -Place orders
  - -Make payments

Application Layer(Backend)

- Built using Node.js / Express
- Handles:
  - Business logic
  - API requests
  - Data validation
  - Communication with database

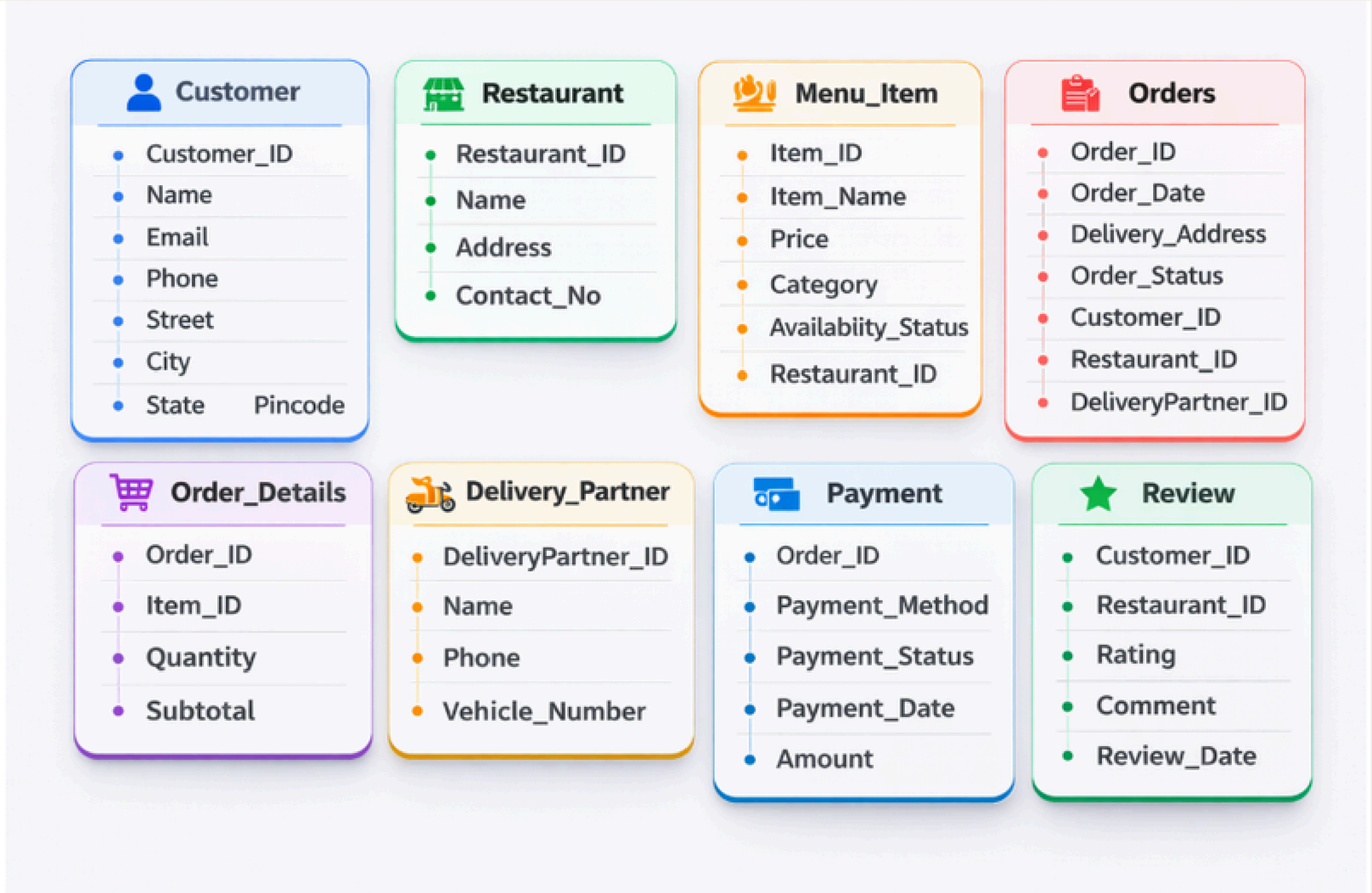
Database Layer (MySQL)

- Stores all system data:
  - Customers
  - Restaurants
  - Orders
  - Menu Items
  - Payments
  - Reviews
- Maintains relationships using foreign keys

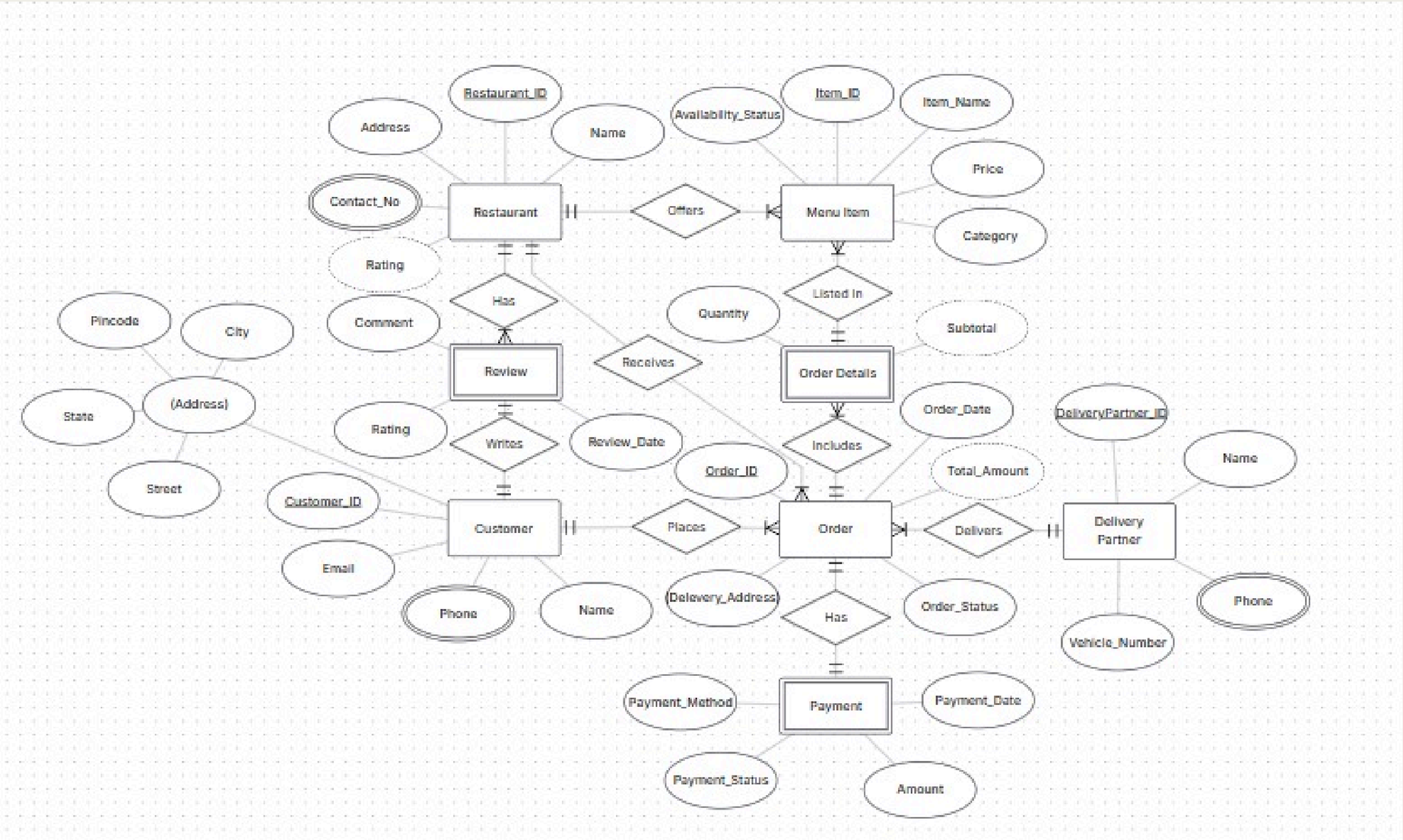


ENTITIES AND ATTRUBUTES

The database consists of eight entities.



E-R DIAGRAM


Restaurant_ID
Address
Name
Availability_Status
Item_ID
Item_Name
Price
Contact_No
Restaurant
Offers
Menu Item
Category
Rating
Has
Listed In
Pincode
City
Comment
Quantity
Subtotal
Review
Receives
Order Details
State
(Address)
Rating
Writes
Review_Date
Includes
Order_Date
DeliveryPartner_ID
Order_ID
Total_Amount
Name
Street
Customer_ID
Customer
Places
Order
Delivers
Delivery Partner
Email
Phone
Name
Delivery_Address
Has
Order_Status
Phone
Vehicle_Number
Payment_Method
Payment
Payment_Date
Payment_Status
Amount

RELATIONSHIPS:

ER Diagram Relationships
in Food Delivery Management System

Customer places Order
One customer can place multiple orders (1:M).

Restaurant offers Menu Items
One restaurant provides many menu items (1:M).

Order includes Menu Items
Many-to-many relationship via Order_Details table (M:N).

Delivery Partner delivers Order
One delivery partner handles multiple orders (1:M).

Customer writes Review
A customer can write multiple reviews (1:M).

Restaurant receives Review
One restaurant can receive multiple reviews (1:M).

Order has Payment
Each order is associated with one payment (1:1).

Customer has Contact
One customer can have multiple phone numbers (1:M).

Delivery Partner has Contact
One delivery partner can have multiple phone numbers (1:M).

Key Observations:

- Most relationships are 1:M
- One M:N relationship resolved using Order_Details
- Payment follows 1:1 relationship
- Contact attributes are multivalued

SQL Implementation

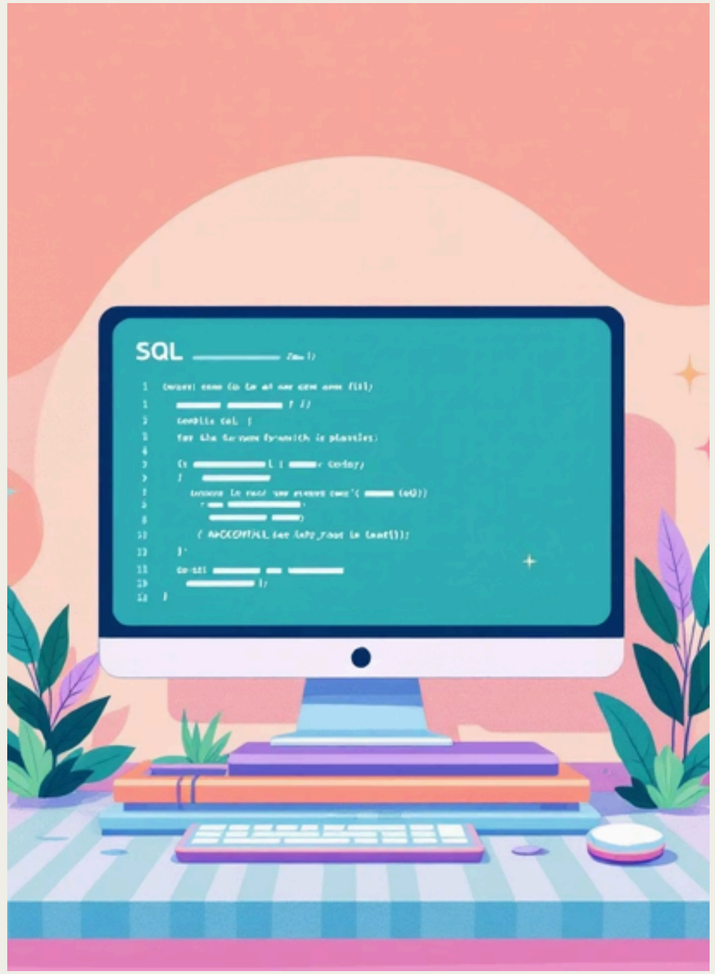
Table Creation:

Implemented with proper constraints, primary keys, and foreign key relationships

```
CREATE TABLE Customer (
  Customer_ID INT PRIMARY
KEY,
  Name VARCHAR(50) NOT
NULL,
  Email VARCHAR(100) UNIQUE
);
```

Example Operations

- INSERT: Add new customers and orders
- SELECT: Query orders and menu items
- UPDATE: Modify order status
- DELETE: Remove cancelled orders



NORMALIZATION

1NF (First Normal Form)

- No multivalued attributes
- Separate tables for contacts
- Example: Customer_Contact, Restaurant_Contact

2NF (Second Normal Form)

- No partial dependency
- All attributes depend on full primary key
- Example: Order_Details (order_id, item_id → quantity)

3NF (Third Normal Form)

- No transitive dependency
- All attributes depend only on primary key
- Example: Customer details depend only on customer_id

CONCLUSION:

Complete Database Degisn

Successfully designed comprehensive ER diagram with all entities, attributes, and relationships

Normalization Achieved

Applied 1NF, 2NF, and 3NF to ensure data integrity and eliminate redundancy

Proper Implementation

Converted ER model into relational schema with appropriate foreign keys and constraints

Real-World Ready

System supports complete food delivery operations from order placement to delivery and payment